



**FACULDADE DE INFORMÁTICA E ADMINISTRAÇÃO
PAULISTA: ENGENHARIA DE DADOS**

Graph Databases and Analytics

SÃO PAULO
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29ABD

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Exercício1 –Construindo grafos

Construir algumas consultas para recomendação de produtos a partir

Import results
Total time: 00:01:58

Run import completed
Import completed successfully.

Time taken	File size	File rows	Nodes created	Properties set	Labels added	Query time
00:00:31	43.5 M/B	541,909	4,070	100,922	4,070	00:00:29

Produto data (3).csv

Time taken	File size	File rows	Nodes created	Properties set	Labels added	Query time
00:00:18	43.5 M/B	541,909	4,372	4,372	4,372	00:00:16

Cliente data (3).csv

Time taken	File size	File rows	Relationships created	Properties set	Query time
00:00:08	43.5 M/B	541,909	4,372	4,372	00:00:06

comprou data (3).csv

```

MATCH(p:Produto) <- [comprou] - (c:Cliente) -> (t:Produto) WHERE p.StockCode = "727867" return p.Description as produto, count(c) as qtdecli, t.Description as comprou, count(e) as qtde, t.StockCode as qtos

```

produto	qtdecli	comprou	qtos	t.StockCode
"FELT EGG COSY CHICKEN"	39	"FELT EGG COSY WHITE RABBIT"	39	"727261"
"FELT EGG COSY CHICKEN"	26	"PACK OF 22 RETROSPOT CAMP CAJIS"	26	"727273"
"FELT EGG COSY CHICKEN"	23	"PAPER CHALK KIT 56'S CHRISTMAS"	23	"727866"
"FELT EGG COSY CHICKEN"	22	"NATURAL SLATE HEART CHALKBOARD"	22	"727457"
"FELT EGG COSY CHICKEN"	22	"GINGERBREAD MAN COOKIE CUTTER"	22	"727864"
"FELT EGG COSY CHICKEN"	21	"HEART OF WICKER LARGE"	21	"727478"
"FELT EGG COSY CHICKEN"	21	"LUNCH BAG CARLS BLUE"	21	"727278"
"FELT EGG COSY CHICKEN"	21	"PAPER CHALK KIT EMPIRE"	21	"727864"
"FELT EGG COSY CHICKEN"	21	"SET OF 3 CAFE TINS PANTRY DESIGN"	21	"727278"
"FELT EGG COSY CHICKEN"	20	"SET OF 4 PANTRY JELLY MOLDING"	20	"727993"

```

MATCH(p:Produto) <- [comprou] - (c:Cliente) -> (t:Produto) WHERE p.StockCode = "72749" and t <- p WITH t, count(DISTINCT c) as clientes return t.Description as produto, clientes ORDER BY clientes desc

```

produto	clientes
"FELTCRAFT PRINCESS LOLA DOLL"	181
"FELTCRAFT PRINCESS OLIVIA DOLL"	133
"PINK CREAM FELT CRAFT TRUNKET BOX"	128

Relationship type
Name: comprou

File: data (3).csv

Node ID mapping

Node	ID	ID column
From: Cliente	CustomerID	CustomerID
To: Produto	StockCode	StockCode

Properties

Name	Type	Column
Quantity	integer	Quantity

a) do produto mais vendido

Database: reddy/movies User: root@

```
root@: MATCH (c:Cliente)-[:comprou]->(p:Produto) WITH p, count(*) AS vendas RETURN p.StockCode AS codigo_produto, p.Description AS produto, vendas ORDER BY vendas DESC LIMIT 1
```

codigo_produto	produto	vendas
"22423"	"REGENCY CAKESTAND 3 TIER"	887

Started streaming 1 record after 265 ms and completed after 948 ms.

b) de um cliente qualquer

```

MATCH (c:Cliente {CustomerID:17650})-[:comprou]->(p:Produto)
MATCH (c2:Cliente)-[:comprou]->(p2:Produto)
WHERE c2 <> c
MATCH (c2)-[:comprou]->(rec:Produto)
WHERE NOT (c2)-[:comprou]->(rec)
WITH rec, count(DISTINCT c2) AS recomendacoes
RETURN rec.StockCode AS codigo_produto,
rec.Description AS produto_recomendado,
recomendacoes
ORDER BY recomendacoes DESC
LIMIT 20
    
```

codigo_produto	produto_recomendado	recomendacoes
"22423"	"REGENCY CAKESTAND 3 TIER"	579
"47566"	"PARTY BUNTING"	496
"84879"	"ASSORTED COLOUR BIRD ORNAMENT"	488
"858998"	"JUMBO BAG RED RETROSPOT"	479
"22728"	"SET OF 3 CAKE TINS PANTRY DESIGN "	439
"22469"	"HEART OF WICKER SMALL"	426
"22457"	"NATURAL SLATE HEART CHALKBOARD "	417
"22886"	"PAPER CHAIN KIT 50'S CHRISTMAS "	415
"23298"	"SPOTTY BUNTING"	411
"21212"	"PACK OF 72 RETROSPOT CAKE CASES"	408

Started streaming 10 records after 52 ms and completed after 5.364 ms.

c) de outro produto qualquer

```
root@: MATCH (p:Produto {StockCode:"85123A"})-[:comprou]-(c:Cliente)-[:comprou]->(t:Produto) WHERE t <> p WITH t, count(DISTINCT c) AS clientes RETURN t.StockCode AS codigo_produto, t.Description AS produto_recomendada, clientes
```

codigo_produto	produto_recomendada	clientes
"21733"	"RED HANGING HEART T-LIGHT HOLDER"	312
"22469"	"HEART OF WICKER SMALL"	266
"22478"	"HEART OF WICKER LARGE"	266
"22423"	"REGENCY CAKESTAND 3 TIER"	262
"47566"	"PARTY BUNTING"	255
"22457"	"NATURAL SLATE HEART CHALKBOARD "	250
"84879"	"ASSORTED COLOUR BIRD ORNAMENT"	238
"22884"	"PINK HANGING HEART T-LIGHT HOLDER"	230
"858998"	"JUMBO BAG RED RETROSPOT"	222
"82482"	"WOODEN PICTURE FRAME WHITE FINISH"	216

Started streaming 10 records after 597 ms and completed after 1.007 ms.

Exercício 2: Case Varejo - Hands on aula 2 - página 14 - responder questões grifadas

Criar uma base de dados a partir do script load_data_transacoes.txt

Após carga verificar o modelo

CALL db.schema.visualization()

The screenshot shows the Neo4j Desktop interface. On the left, the 'Database Information' panel displays nodes (27,150) and relationships (975,000). The central graph visualization shows a network of nodes including 'item', 'category', 'customer', and 'transaction'. On the right, the 'Results overview' panel shows a summary of the data. At the bottom, a Cypher query is executed:

```

neo4j CALL db.schema.visualization()
CREATE INDEX IF NOT EXISTS FOR (i:item) ON i StockCode;
CREATE INDEX IF NOT EXISTS FOR (c:customer) ON c CustomerID;
CREATE INDEX IF NOT EXISTS FOR (t:transaction) ON t TransactionID;
CREATE INDEX IF NOT EXISTS FOR (c:category) ON c Category;
  
```

1) Explorar os dados respondendo:

Total de itens, de categorias, de clientes, de países e de transações

The screenshot shows the Neo4j Desktop interface with a Cypher query and its results. The query is:

```

MATCH (i:item)
WITH count(i) AS total_items
MATCH (c:category)
WITH total_items, count(c) AS total_categorias
MATCH (cu:customer)
WITH total_items, total_categorias, count(cu) AS total_clientes
MATCH (co:country)
WITH total_items, total_categorias, total_clientes, count(co) AS total_paises
RETURN total_items,
       total_categorias,
       total_clientes,
       total_paises,
       count(t) AS total_transacoes
  
```

The results are displayed in a table:

total_items	total_categorias	total_clientes	total_paises	total_transacoes
2864	12	4372	38	19849



Quais os dez clientes com o maior número de transações e quais os produtos mais comprados?

```
neo4j$ MATCH (c:Customer)-[:WAS_TRANSACTION]->(t:Transaction)
WITH c, count(t) AS total_transacoes
RETURN c.customer_id AS cliente,
total_transacoes
ORDER BY total_transacoes DESC
LIMIT 10
```

cliente	total_transacoes
12748	236
14911	225
17841	145
13889	132
15311	121
14686	120
12071	113
14646	102
16029	91
13488	89

Started streaming 10 records after 103 ms and completed after 188 ms.

```
neo4j$ MATCH (c:Customer)-[:BOUGHT]->(i:Item) WITH i, sum(b.Quantity) AS total_vendas
RETURN i.stock_code AS codigo_produto, i.description AS produto, total_vendas
ORDER BY total_vendas DESC LIMIT 10
```

codigo_produto	produto	total_vendas
23843	"PAPER CRAFT LITTLE BIRDIE"	88995
23166	"MEDIUM CERAMIC TOP STORAGE JAR"	78833
22197	"POPCORN HOLDER"	56921
84877	"WORLD WAR 2 GLIDERS ASSTO DESIGNS"	55847
84879	"ASSORTED COLOUR BIRD ORNAMENT"	36461
21212	"PACK OF 72 RETROSPOT CAKE CASES"	36419
23884	"RABBIT NIGHT LIGHT"	38788
22492	"MINT PAINT SET VINTAGE "	26633
22616	"PACK OF 12 LONDON TISSUES "	26135
21977	"PACK OF 60 PINK PAISLEY CAKE CASES"	24854

Started streaming 10 records after 84 ms and completed after 3,021 ms.

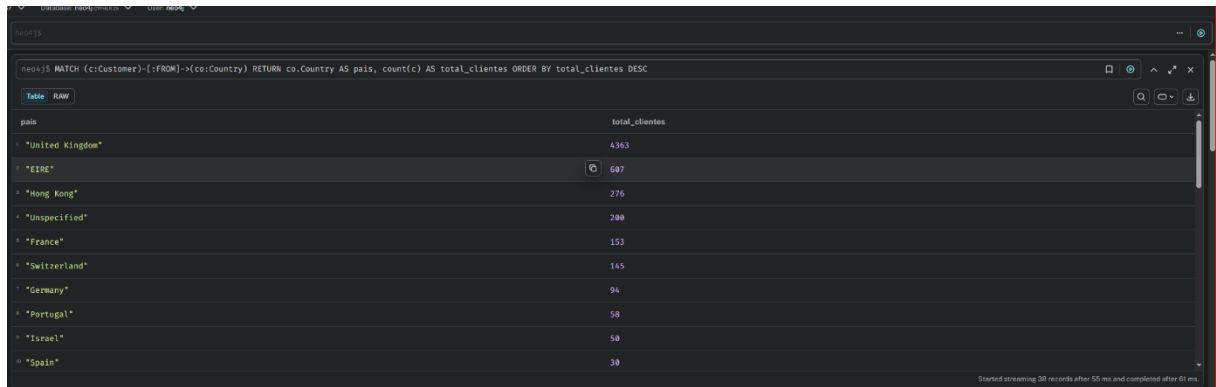
Total de itens por categoria

```
neo4j$ MATCH (i:Item)-[:TYPE]->(c:Category) RETURN c.category AS categoria, count(i) AS total_items
ORDER BY total_items DESC
```

categoria	total_items
"HOME DÉCOR"	866
"KITCHEN"	593
"STATIONERY"	312
"TOY"	244
"ACCESSORIES"	214
"SEASONAL"	202
"GARDEN"	80
"PARTY"	79
"JEWELRY"	68
"ARTS AND CRAFTS"	66

Started streaming 12 records after 81 ms and completed after 92 ms.

Total de clientes por país



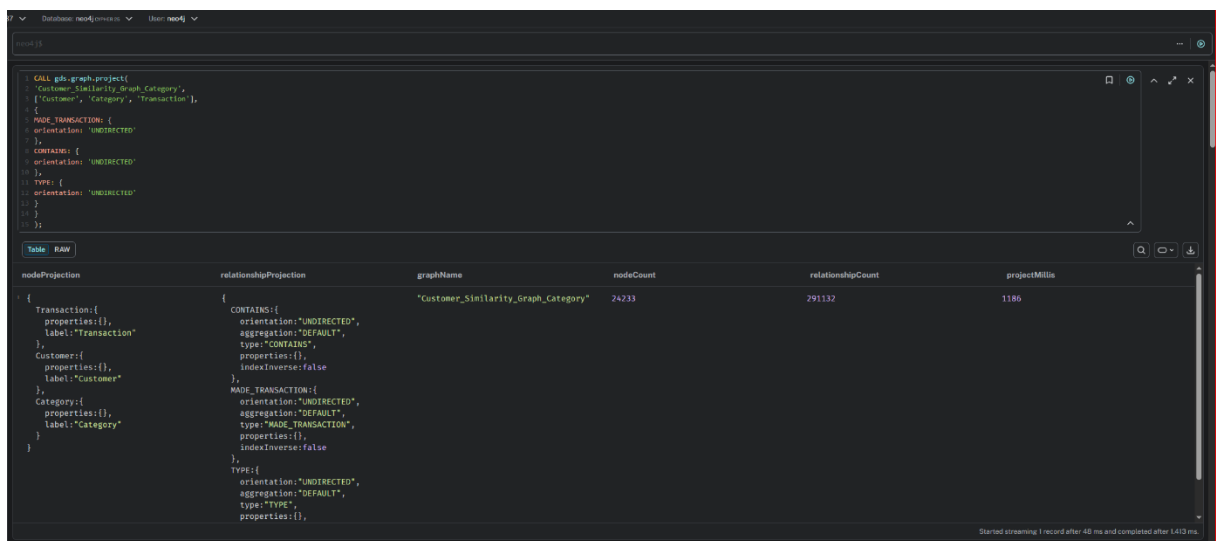
```

neo> MATCH (c:Customer)-[:FROM]->(co:Country) RETURN co.country AS pais, count(c) AS total_clientes ORDER BY total_clientes DESC
    
```

pais	total_clientes
"United Kingdom"	4363
"EIRE"	697
"Hong Kong"	276
"Unspecified"	200
"France"	153
"Switzerland"	145
"Germany"	94
"Portugal"	58
"Israel"	58
"Spain"	39

2. Realizar Segmentação de Clientes

- Similaridade por Cliente por Categoria – criar projeção Customer_Similarity_Graph_Category

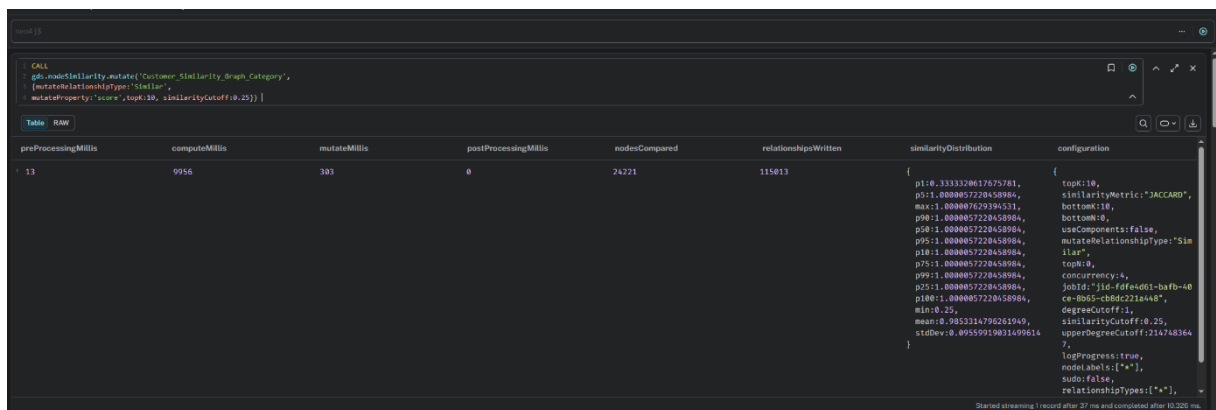


```

CALL gds.graph-project(
  'Customer_Similarity_Graph_Category',
  ['Customer', 'Category', 'Transaction'],
  [
    {
      MADE_TRANSACTION: {
        orientation: 'UNDIRECTED'
      },
      CONTAINS: {
        orientation: 'UNDIRECTED'
      },
      TYPE: {
        orientation: 'UNDIRECTED'
      }
    }
  ]
)
    
```

nodeProjection	relationshipProjection	graphName	nodeCount	relationshipCount	projectMillis
{ Transaction: { properties: {}, label: "Transaction" }, Customer: { properties: {}, label: "Customer" }, Category: { properties: {}, label: "Category" } }	{ CONTAINS: { orientation: "UNDIRECTED", segregation: "DEFAULT", type: "CONTAINS", properties: {}, indexInverse: false }, MADE_TRANSACTION: { orientation: "UNDIRECTED", segregation: "DEFAULT", type: "MADE_TRANSACTION", properties: {}, indexInverse: false }, TYPE: { orientation: "UNDIRECTED", segregation: "DEFAULT", type: "TYPE", properties: {} } }	"Customer_Similarity_Graph_Category"	24233	291132	1186

- Similaridade por Cliente por Categoria – gravando relacionamento Similar



```

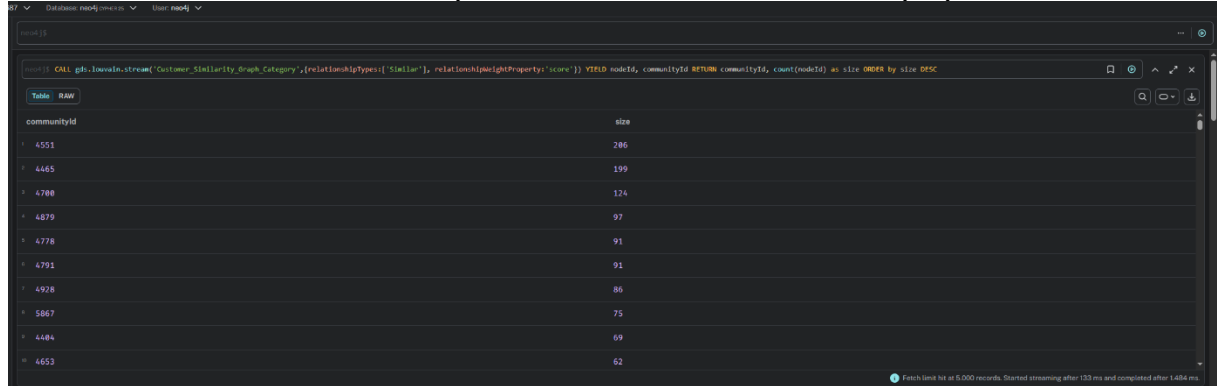
CALL
  gds.nodeSimilarity.write('Customer_Similarity_Graph_Category',
    [writeRelationshipType: 'Similar',
      writeProperty: 'score', 'topk:10', similarityCutoff: 0.25]
  )
    
```

preProcessingMillis	computeMillis	mutateMillis	postProcessingMillis	nodesCompared	relationshipsWritten	similarityDistribution	configuration
13	9956	303	0	24221	115813	{ p10: 333320617075781, p51: 0.000057220458984, max: 1.000007629394331, p90: 1.0000057220458984, p50: 1.0000057220458984, p95: 1.0000057220458984, p10: 1.0000057220458984, p75: 1.0000057220458984, p99: 1.0000057220458984, p25: 1.0000057220458984, p100: 1.0000057220458984, min: 0.25, mean: 0.9453316796261949, stdDev: 0.09559919831699614 }	{ topk: 10, similarityMetric: "JACCARD", bottomN: 0, bottomM: 0, useComponents: false, mutateRelationshipType: "Similar", topk: 0, concurrency: 4, jobId: "316-fdfe4d01-bafb--0ce-8b65-cbdc221a448", degreeCutoff: 1, similarityCutoff: 0.25, upperDegreeCutoff: 2147483647, logProgress: true, nodeLabels: ["*"], sudo: false, relationshipTypes: ["*"] }

2) Explicar o comando mutate e destacar os dez clientes relacionados

O comando Mutate vai gravar o resultado temporariamente na projeção de grafo, mas sem persistir os dados diretamente no banco.

c. Louvain comunidade formada a partir do relacionamento Similar e propriedade score

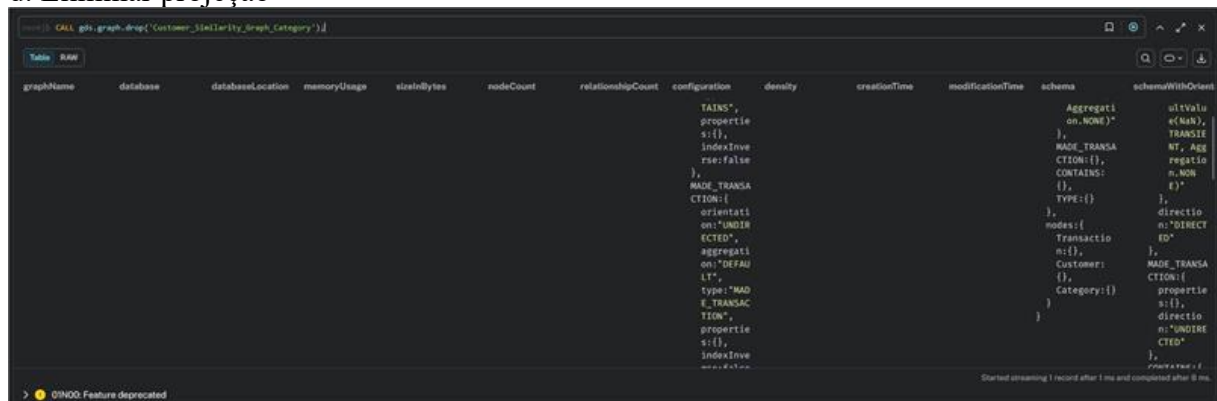


```
CALL gds.louvain.stream('Customer_Similarity_Graph_Category', [relationshipTypes: ['Similar'], relationshipWeightProperty: 'score']) YIELD nodeId, communityId RETURN communityId, count(nodeId) as size ORDER BY size DESC
```

communityId	size
4551	286
4465	199
4788	124
4879	97
4778	91
4791	91
4928	86
5867	75
4484	69
4653	62

3) Quantas comunidades foram criadas com o total de clientes em cada uma?

d. Eliminar projeção



```
CALL gds.graph.drop('Customer_Similarity_Graph_Category')
```

graphName	database	databaseLocation	memoryUsage	sizeInBytes	nodeCount	relationshipCount	configuration	density	creationTime	modificationTime	schema	schemaWithOrient
Customer_Similarity_Graph_Category	default	default	100MB	100MB	7238	352268	{ "TAINS": { "properties": { "s": { "indexInverse": false } }, "MADE_TRANSACTION": { "orientation": "UNDIRECTED", "aggregation": "DEFAULT", "type": "BROUGHT", "properties": { "s": { "indexInverse": false } } } } }	0.000000	2023-10-27 10:00:00	2023-10-27 10:00:00	{ "Aggregation": "NONE", "MADE_TRANSACTION": { "type": "BROUGHT", "properties": { "s": { "indexInverse": false } } } }	{ "type": "BROUGHT", "properties": { "s": { "indexInverse": false } } }

e. Cliente similaridade por item



```
CALL gds.graph.project(
  'Customer_Similarity_Graph_Items',
  [
    {
      type: 'Customer',
      label: 'Customer'
    },
    {
      type: 'Item',
      label: 'Item'
    }
  ],
  {
    'BOUGHT': {
      orientation: 'NATURAL',
      aggregation: 'DEFAULT',
      type: 'BROUGHT',
      properties: {
        's': {
          indexInverse: false
        }
      }
    }
  }
)
```

nodeProjection	relationshipProjection	graphName	nodeCount	relationshipCount	projectMillis
{ Customer: { properties: {}, label: 'Customer' }, Item: { properties: {}, label: 'Item' } }	{ BOUGHT: { orientation: 'NATURAL', aggregation: 'DEFAULT', type: 'BROUGHT', properties: { s: { indexInverse: false } } } }	Customer_Similarity_Graph_Items	7238	352268	75

f. Node similaridade

```
CALL gds.nodeSimilarity.mutate(
  'Customer_Similarity_Graph_Items',
  {
    mutateRelationshipType: 'Similar',
    mutateProperty: 'score',
    topK: 10,
    similarityCutoff: 0.1
  }
);
```

preProcessingMillis	computeMillis	mutateMillis	postProcessingMillis	nodesCompared	relationshipsWritten	similarityDistribution	configuration
0	3168	16	0	4371	17298	{ p1:0.09999990463256836, p5:0.10119009017944336, max:0.50880221083496893, p90:0.10959047317584083, p50:0.12209272384643555, p95:0.18762445449829102, p10:0.10285711288452148, p75:0.14328336715698242, p99:0.2288802084675793, p25:0.1088785062066211, p100:0.5088021831512451, min:0.09999990463256836, 6, }	{ topK:10, similarityMetric:"JACCA RD", bottom:10, bottom:0, useComponents:false, mutateRelationshipType:"Similar", topK:0, concurrency:4, jobId:"jid-24c3fb79-f90e-44e4-b981-81e0f1237684", degreeCutoff:1, similarityCutoff:0.1, upperDegreeCutoff:2147483647, logProgress:true, nodeLabels:["*"], }

Started streaming 1 record after 1 ms and completed after 3.107 ms.

4. Destacar os dez itens relacionados

g. Louvain

```
CALL gds.louvain.stream(
  'Customer_Similarity_Graph_Items',
  { relationshipTypes: ['Similar'], relationshipWeightProperty: 'score' }
)
YIELD nodeId, communityId
RETURN communityId, count(nodeId) AS size
ORDER BY size DESC;
```

communityId	size
4154	114
3321	31
4713	12
4229	6
4529	5
5062	5
6394	4
3642	4
4089	4
4601	4

Started streaming 4,503 records after 1 ms and completed after 1.103 ms.

h. Louvain Mutate

```
CALL gds.louvain.mutate(
  'Customer_Similarity_Graph_Items',
  {
    relationshipTypes: ['Similar'],
    relationshipWeightProperty: 'score',
    mutateProperty: 'louvainCommunity'
  }
);
```

modularity	modularities	ranLevels	communityCount	communityDistribution	preProcessingMillis	computeMillis	postProcessingMillis	mutateMillis	nodePropertiesWritten	configuration
				{ 9687873, p999:1334 }						{ 277-c999-453a-815d-766931deb9fe", forceSeedOptimization:false, logProgress:true, includeIntermediateCommunities:false, nodeLabels:["*"], sudo:false, relationshipTypes:["Similar"], mutateProperty:"louvainCommunity", tolerance:0.0001 }

Started streaming 1 record after 1 ms and completed after 1.296 ms.

5) Quantas comunidades foram criadas com o total de itens em cada uma?

```
CALL gds.invest.stream(
  'Customer_Similarity_Graph_Items',
  { relationshipTypes: ['Similar'], relationshipWeightProperty: 'score' }
);
WITH nodeId, communityId
RETURN communityId, count(nodeId) AS size
ORDER BY size DESC;
```

communityId	size
2929	638
3395	559
4465	395
3898	367
4154	258
6789	127
6997	123
4575	56
6391	32
4229	6

```
CALL gds.graph.writeNodeProperties(
  'Customer_Similarity_Graph_Items',
  ['louvainCommunity']
);
```

i. Eliminar a projeção

```
neo4j CALL gds.graph.drop('Customer_Similarity_Graph_Items');
```

graphName	database	databaseLocation	memoryUsage	sizeInBytes	nodeCount	relationshipCount	configuration	density	creationTime	modificationTime	schema	schemaWithOrient
"Customer_Similarity_Graph_Items"	"neo4j"	"local"	""	-1	7236	369566	{ relationship rejection: { BOUGHT: { orientation: "NATUR", type: "BOUGHT", property: { s: {}, indexInverse: false } }, jobId: "job-2198e190-1b60-4"	0.087059190723797889	2026-04-08T02:00:00Z [UTC]	2026-04-08T02:00:00Z [UTC]	{ graphProperties: {}, relationship s: { Similar: { score: "FL", default: (Default), value: (NaN), transient: { BOUGHT: { aggregation: { ANTIEN, Aggregation: (NaN) } } }, nodes: { Items: { louvainCommunity: { directio	{ graphProperties: {}, relationship s: { Similar: { score: "FL", default: (Default), value: (NaN), transient: { BOUGHT: { aggregation: { ANTIEN, Aggregation: (NaN) } } }, nodes: { Items: { louvainCommunity: { directio

3) Item Similaridade

a. Item similaridade = 6115 (escolher comunidade existente em clientes)

```
CALL gds.graph.project.cypher(
  'item_similarity_3653',
  'MATCH (i:Item)->-(c:Customer)
  WHERE c.louvainCommunity = 3653
  RETURN DISTINCT id(i) AS id',
  'MATCH (i:Item)->-(c:Customer)->-(i2:Item)
  WHERE c.louvainCommunity = 3653
  AND id(i1) < id(i2)
  RETURN id(i1) AS source, id(i2) AS target, count(c) AS weight'
);
```

nodeQuery	relationshipQuery	graphName	nodeCount	relationshipCount	projectMillis
"MATCH (i:Item)->-(c:Customer) WHERE c.louvainCommunity = 3653 RETURN DISTINCT id(i) AS id"	"MATCH (i:Item)->-(c:Customer)->-(i2:Item) WHERE c.louvainCommunity = 3653 AND id(i1) < id(i2) RETURN id(i1) AS source, id(i2) AS target, count(c) AS weight"	"item_similarity_3653"	2755	282041	26965

b. Gravar relacionamento similaridade

```
CALL gds.similarity.write(
  'item_copurchase_3653',
  {
    writeRelationshipType: 'SIMILAR_3653',
    writeProperty: 'score',
    topK: 10,
    similarityCutoff: 0.1
  }
);
```

preProcessingMillis	computeMillis	writeMillis	postProcessingMillis	nodesCompared	relationshipsWritten	similarityDistribution	configuration
0	6231	260	0	2754	27306	{ p91:0.34394087815551758, p5:0.33333149227805273, max:1.000007629394531, p98:0.938239574432373, p58:0.8838725890826855, p95:0.9474787712097108, p18:0.5249452236822849, p75:0.928891459655762, p99:0.9685626829968262, p25:0.8000025749286543, p100:1.000007152557373, min:0.099999946325683, mean:0.889199505286069, stdDev:0.18674354952130 }	{ topK:10, writeConcurrency:4, similarityMetric:'JACCA RD', bottomK:10, bottomN:0, useComponents:false, topN:0, concurrency:4, jobId:'jid-5465b0c1-3a8 f-4096-99e5-6e3d673156d c', writeProperty:'score', degreeCutoff:1, writeRelationshipTyp e:'SIMILAR_3653', similarityCutoff:0.1, upperDegreeCutoff:11474 83647, }

Started streaming 1 record after 10 ms and completed after 6304 ms.

c. Item closeness

```
CALL gds.closeness.write(
  'item_copurchase_3653',
  { writeProperty: 'Closeness_Centrality_Community_3653' }
);
```

nodePropertiesWritten	preProcessingMillis	computeMillis	postProcessingMillis	writeMillis	centralityDistribution	configuration
2755	0	258	1	19	{ p99:0.9659786224365234, min:0.0, max:1.000007629394531, p98:0.9325847625732422, mean:0.8169444829942809, p999:1.0000092218456984, p50:0.8542957305980203, p95:0.9465887781982422, p75:0.9838608921630859 }	{ jobId:'jid-3246b0cf-2e8d-4e 75-8816-1209c291578c', writeConcurrency:4, writeProperty:'Closeness_Ce ntrality_Community_3653', useHadoopFault:false, logProgress:true, nodeLabels:['*'], sudo:false, relationshipTypes:['*'], writeToResultStore:false, concurrency:4 }

Started streaming 1 record after 7 ms and completed after 290 ms.

d. Item Pagerank

```
CALL gds.pagerank.write(
  'item_copurchase_3653',
  {
    relationshipWeightProperty: 'weight',
    writeProperty: 'PageRank_Community_3653'
  }
);
```

iterations	didConverge	centralityDistribution	preProcessingMillis	computeMillis	postProcessingMillis	writeMillis	nodePropertiesWritten	configuration
20	false	{ p99:6.73871193695068 4, min:0.14999961853827 344, max:135.473632812499 97, p98:1.82973842620849 6, mean:0.7213478982078 889, p999:62.749518765875 684, p50:0.23922157287597 656, p95:1.95838832855224 6, p75:0.46930627088056 84 }	0	229	6	16	2755	{ iterations:20, writeConcurrency:4, relationshipWeightPr operty:'weight', concurrency:4, jobId:'jid-2289c9f 2-6d0b-4b54-998b-a9c 184c73b6d', sourceNodes:[], writeProperty:'PageR ank_Community_3653', scaler:'NONE', logProgress:true, nodeLabels:['*'], sudo:false, dampingFactor:0.85, relationshipTypes: ['*'], writeToResultStore:f }

Started streaming 1 record after 7 ms and completed after 268 ms.



Neo4j Cypher query interface showing a table view of items with PageRank and Closeness scores. The query is: `MATCH (i:item)--(c:Customer) WHERE c.LouvainCommunity = 3653 RETURN DISTINCT i.Description AS Item, i.PageRank_Community_3653 AS PageRank, i.Closeness_Centrality_Community_3653 AS Closeness LIMIT 10;`

Item	PageRank	Closeness
"FIRST CLASS LUGGAGE TAG "	0.1529782443588273	0.8169814884587842
"KEY FOB BACK DOOR "	0.1689528142487669	0.9583333333333334
"KEY FOB FRONT DOOR "	0.16882568874372385	0.96875
"KEY FOB GARAGE DESIGN"	0.16186171288328674	0.9698723649484536
"KEY FOB SHED"	0.1633280778239413	0.9693877551828488
"PINK CREAM FELT CRAFT TRINKET BOX "	0.16287568867948827	0.928
"SET OF 4 KNICK KNACK TINS DOTLEY "	0.1673485788238263	0.9885524861878453
"VIP PASSPORT COVER "	0.16137186769133527	0.8766519823788547
"6 RIBBONS RUSTIC CHARM"	0.18872888938169846	0.9439252336448598
"EMBROIDERED RIBBON REEL EMILY "	0.162252184656317	0.8729588196721312

Neo4j Cypher query interface showing a table view of items with scores. The query is: `MATCH p=[(i:item)-[r:SIMILAR_3653]-i2] WHERE r.score > 0.1 AND r.score < 0.8 RETURN i1.Description, i2.Description, r.score AS score ORDER BY score DESC;`

i1.Description	i2.Description	score
"MIRRORED WALL ART PHOTO FRAMES"	"HEART T-LIGHT HOLDER"	0.7998472116119175
"CERAMIC CAKE BOWL + HANGING CAKES"	"CAKE STAND WHITE TWO TIER LACE"	0.799837925445785
"COOKING SET RETROSPOT"	"COFFEE MUG PINK PAISLEY DESIGN"	0.7998266897746967
"COOKING SET RETROSPOT"	"COSY HOUR CIGAR BOX MATCHES "	0.7998266897746967
"COOKING SET RETROSPOT"	"CORDIAL GLASS JUG"	0.7998212689981698
"ANGEL DECORATION PAINTED ZINC "	"4 WILDFLOWER BOTANICAL CANDLES"	0.7998051631758483
"SET OF 4 DIAMOND NAPKIN RINGS"	"SET OF 2 TINS JARDIN DE PROVENCE"	0.7997724687144482
"LANDMARK FRAME CAMDEN TOWN "	"LANDMARK FRAME OXFORD STREET"	0.799752781211372
"WRAP MAGIC FOREST "	"WRAP DOLLY GIRL"	0.7997876823391813
"I LOVE LONDON WALL ART"	"LARGE CAKE STAND HANGING HEARTS"	0.7995955838262273

Encontrar similares

Neo4j Cypher query interface showing a table view of similar items. The query is: `neo4j> CALL gds.nodeSimilarity.stream('item_copurchase_3653', { topK: 1 }) YIELD node1, node2, similarity RETURN gds.util.asNode(node1).Description AS Produto1, gds.util.asNode(node2).Description AS Produto2, similarity;`

Produto1	Produto2	similarity
"WHITE BEADED GARLAND STRING 20LIGHT"	"WRAP WEDDING DAY"	1.0
"WRAP WEDDING DAY"	"WHITE BEADED GARLAND STRING 20LIGHT"	1.0
"LARGE BLACK DIAMANTE HAIRSLIDE"	"MONTANA DIAMOND CLUSTER EARRINGS"	0.9988662131519275
"MONTANA DIAMOND CLUSTER EARRINGS"	"LARGE BLACK DIAMANTE HAIRSLIDE"	0.9988662131519275
"NUMBER TILE VINTAGE FONT 7"	"NUMBER TILE VINTAGE FONT No "	0.9977238239757288
"NUMBER TILE VINTAGE FONT No "	"NUMBER TILE VINTAGE FONT 7"	0.9977238239757288
"COPPER AND BRASS BAG CHARM"	"CRYSTAL SEA HORSE PHONE CHARM"	0.9949748743718593
"CRYSTAL SEA HORSE PHONE CHARM"	"COPPER AND BRASS BAG CHARM"	0.9949748743718593
"CROCHET BEAR RED/BLUE KEYRING"	"CROCHET DOG KEYRING"	0.994535519125683
"CROCHET DOG KEYRING"	"CROCHET BEAR RED/BLUE KEYRING"	0.994535519125683

